

dotenv in Node.js Project

In a Node.js project, **dotenv** is a popular package used to manage **environment variables**.

It allows you to keep sensitive information (like database credentials, API keys, ports, etc.) outside your source code and load them from a `.env` file.

■ Steps to Use dotenv in a Node.js Project

1. Install dotenv

```
npm install dotenv
```

2. Create a `.env` file

In your project root directory, create a file named `.env` and add your environment variables:

```
PORT=5000
DB_HOST=localhost
DB_USER=admin
DB_PASS=secret123
```

■■ Never commit `.env` files to GitHub. Add `.env` to `.gitignore`.

3. Load dotenv in your project

At the very top of your `server.js` / `app.js` file:

```
require("dotenv").config();

const express = require("express");
const app = express();

const PORT = process.env.PORT || 3000;

app.get("/", (req, res) => {
  res.send("Server running with dotenv!");
});

app.listen(PORT, () => {
```

```
console.log(`Server running on http://localhost:${PORT}`);  
});
```

4. Access variables with `process.env`

```
console.log(process.env.DB_HOST); // localhost  
console.log(process.env.DB_USER); // admin  
console.log(process.env.DB_PASS); // secret123
```

■ Best Practices

- Always add `.env` to `.gitignore`.
- Use **default values** with `||` in case variables are missing.
- For production, set environment variables in your **server configuration** (not via `.env`).

■ Is `.env` added to `.gitignore` by default?

No — **`.env` is not added to `.gitignore` by default**.

When you create a Node.js project (using `npm init` or `npx express-generator`), there is usually **no `.gitignore` file created automatically** unless you use a framework or boilerplate that ships with one (like `create-react-app` or `next.js`).

So, if you make a `.env` file yourself, you need to manually add it to `.gitignore`:

Ignore environment variables

```
.env
```

```
.env.local
```

```
.env*.local
```

■ This prevents your sensitive data (like API keys, DB passwords, etc.) from being pushed to GitHub or shared with others.

■■ If you've already committed `.env` to Git, you'll need to remove it from history using:

```
git rm --cached .env
```

```
git commit -m "Remove .env from repo"
```